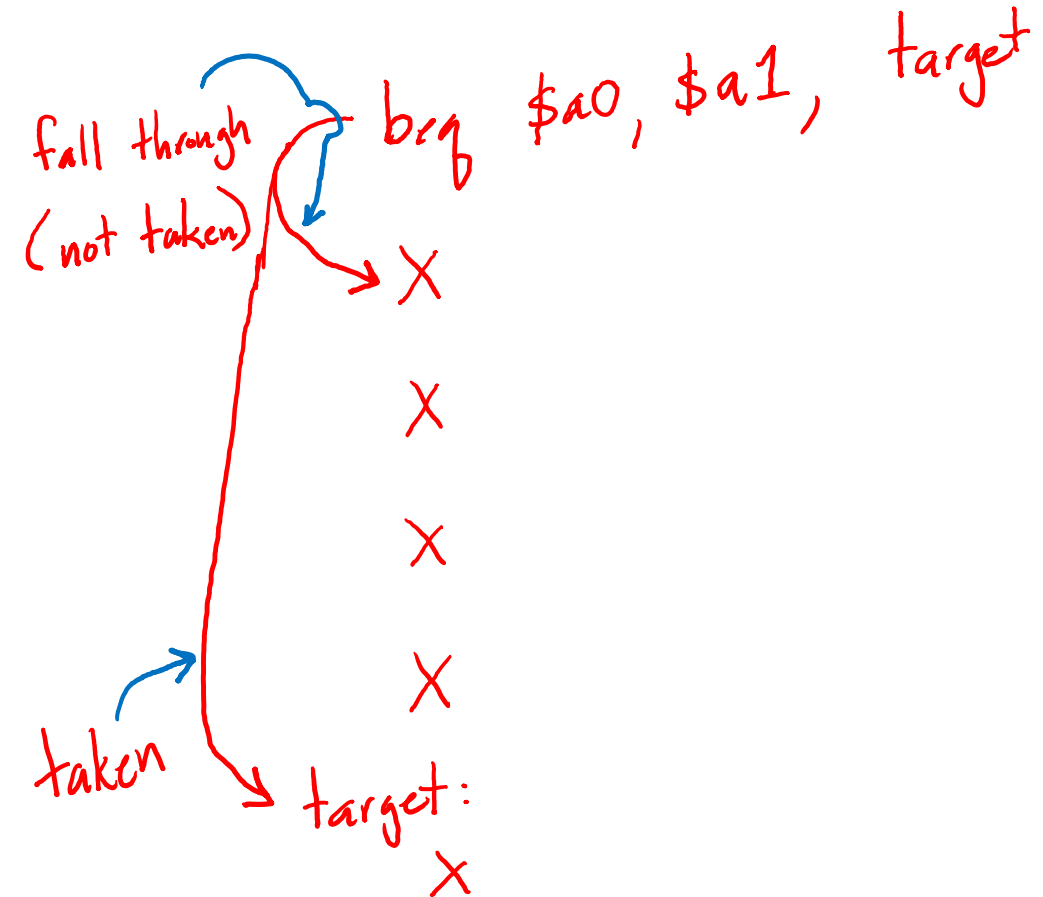
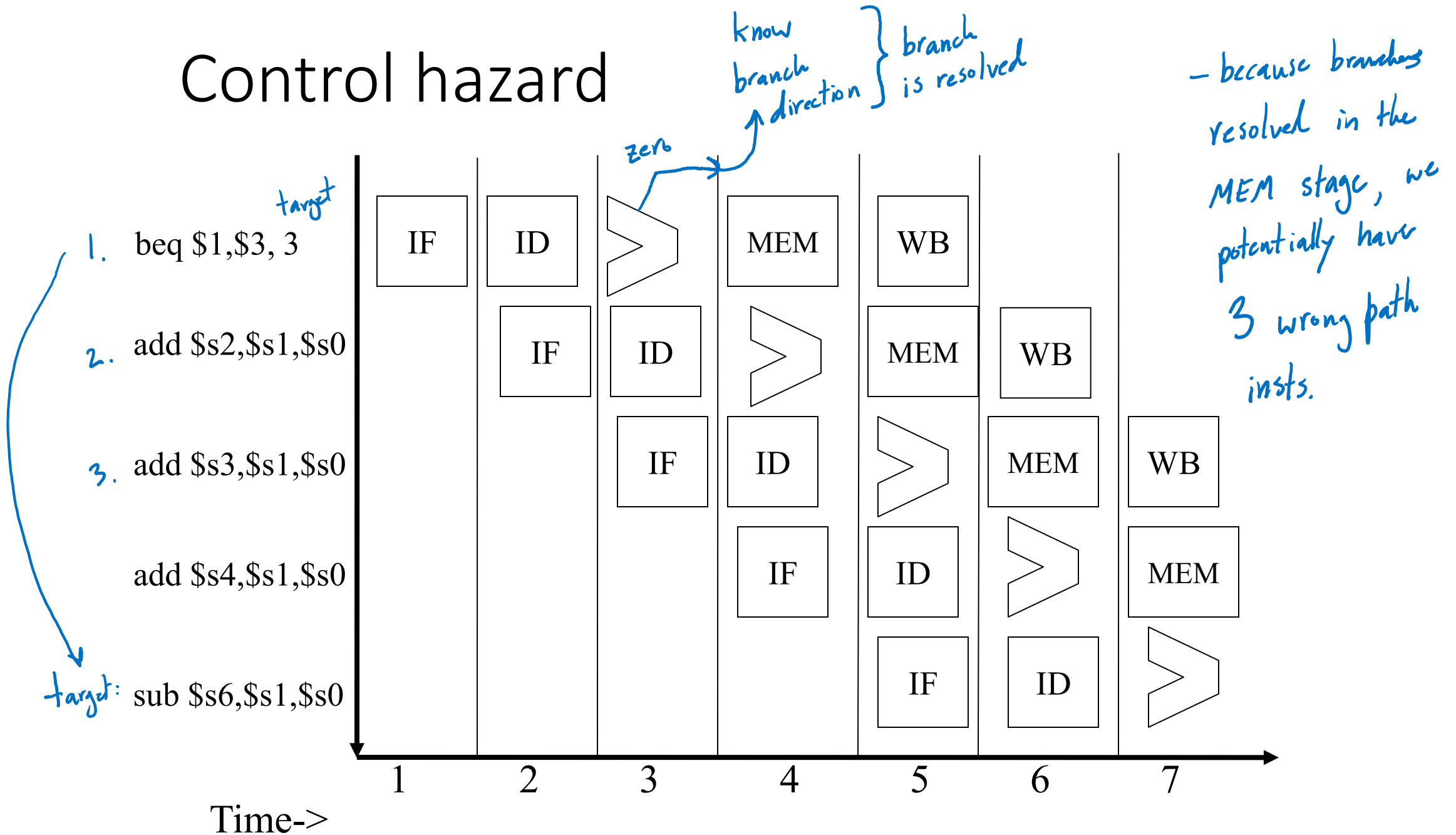


Control (Branch) Hazards



Control hazard



Control hazard techniques

1. Assume that branches are not taken

we run branch

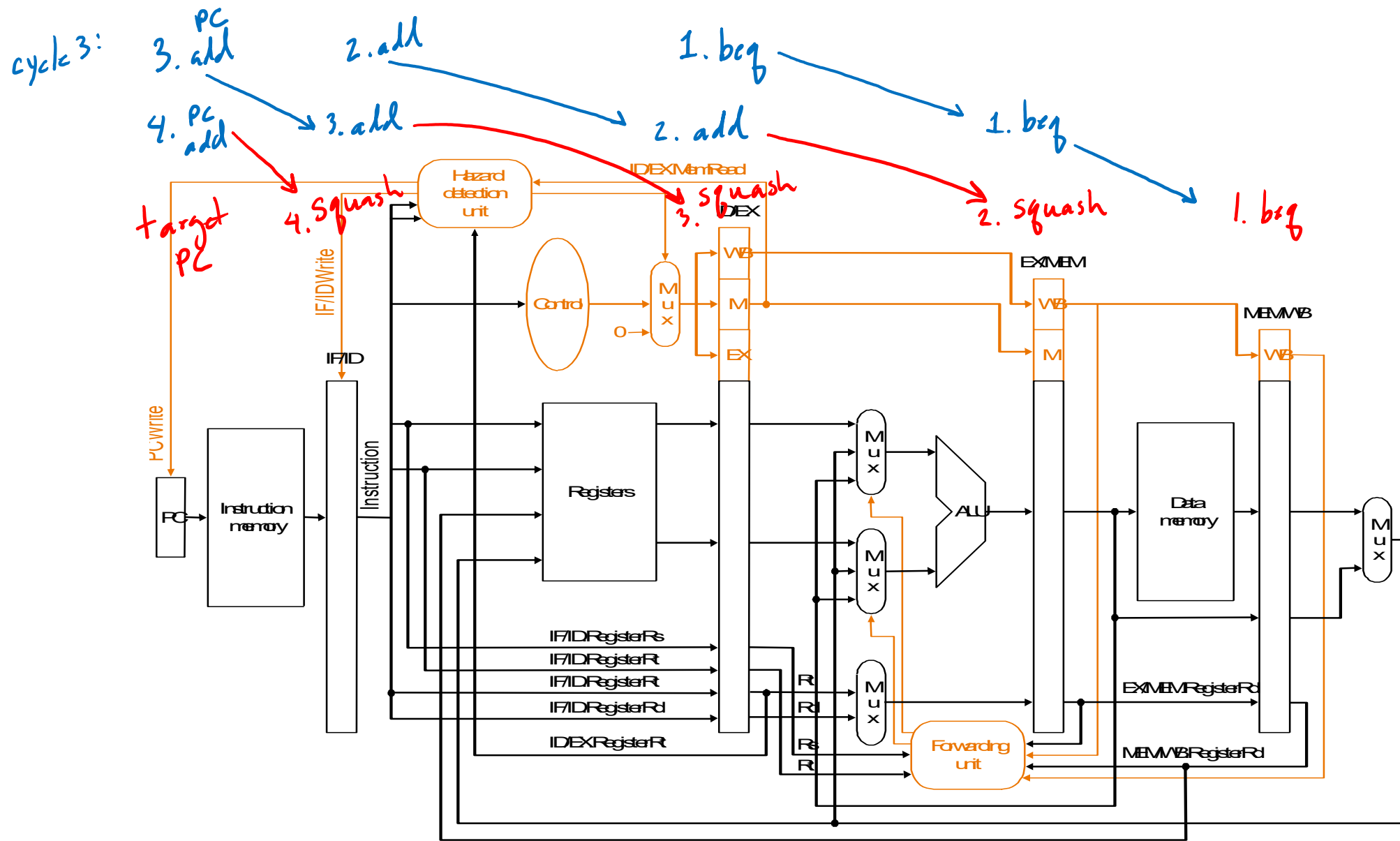
- if the branch is actually not taken,
everything works fine

- if the branch is taken, then turn the

3 wrong path insts. into no-ops
(squash)

→ clear out the
control signals
for those insts.

set to
0



2. Delayed branches

loop: A
B
C
D
E
F
beg ..., ..., loop

CPU must support delayed branching

— as a programmer, you have to fill all the delay slot

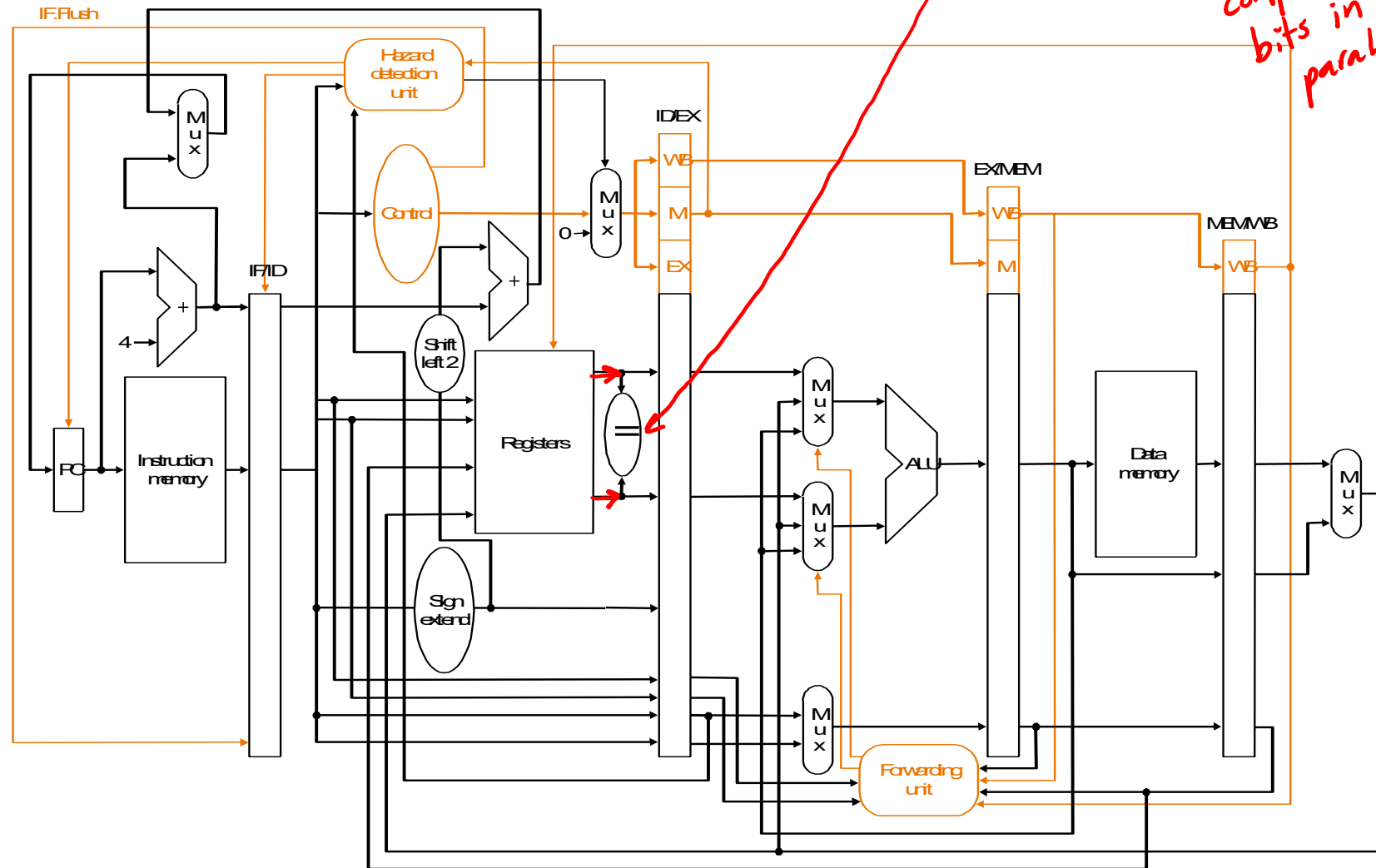
loop: A
B
C

beg ..., ..., loop

D
E
F

a CPU with delayed branches always runs the delay slot insts (this CPU has 3 delay slots)

3. Execute branch earlier



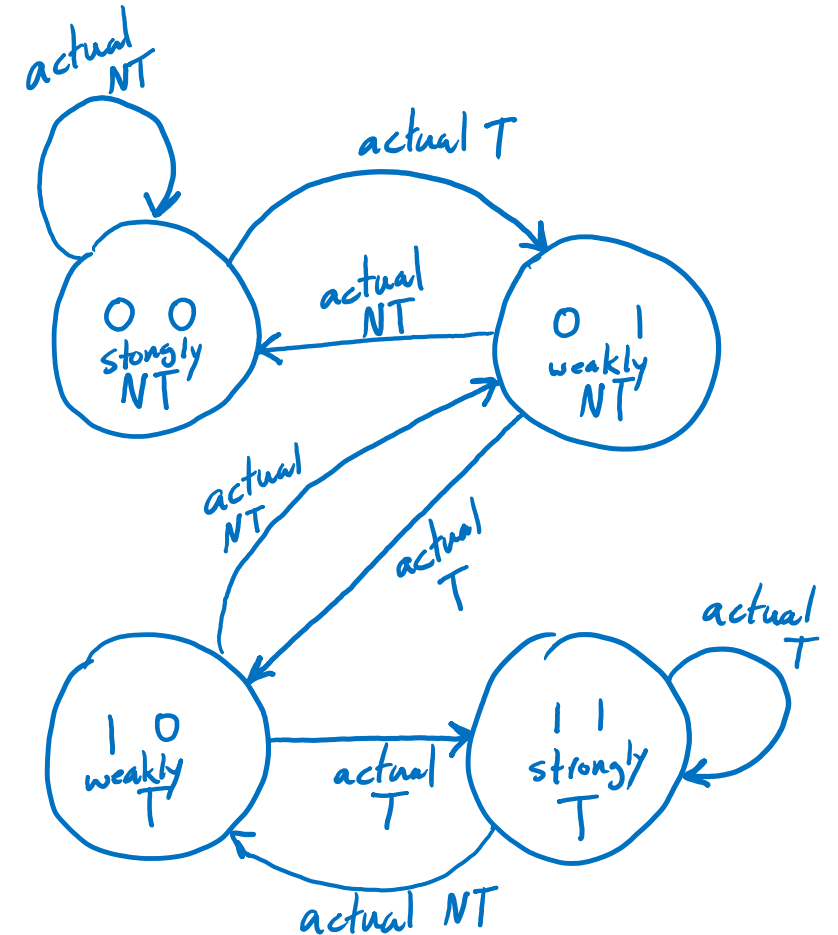
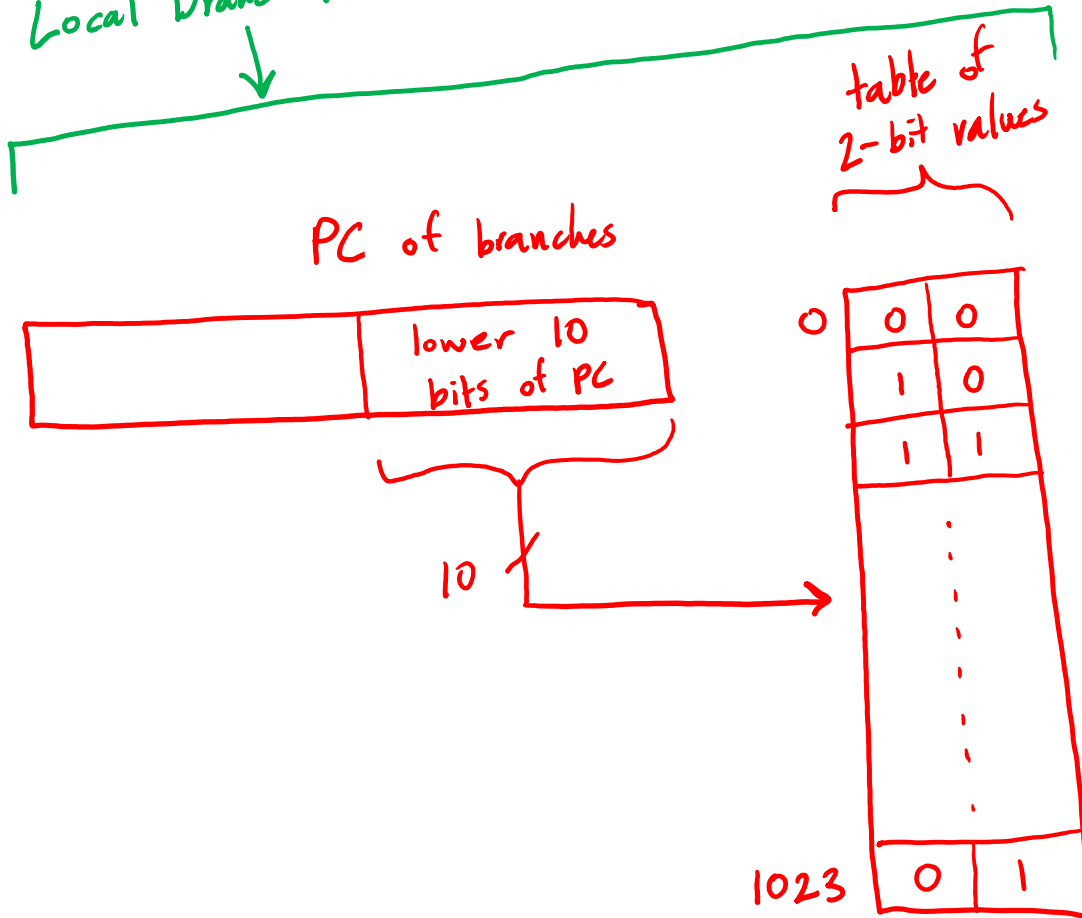
Comparator - fast unit to compare bits in parallel

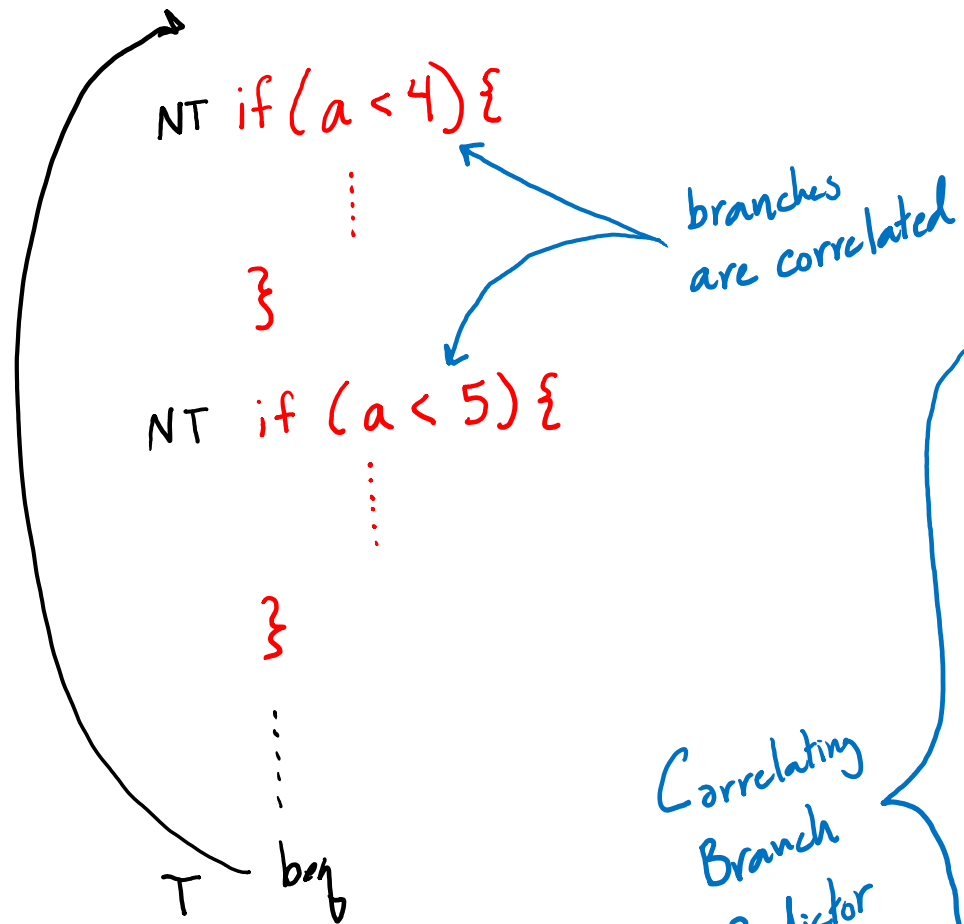
- only need to squash 1 inst. on a taken branch

4. Modern solution

Branch Prediction - while the program is run, a branch predictor predicts branch direction and fetches inst. accordingly

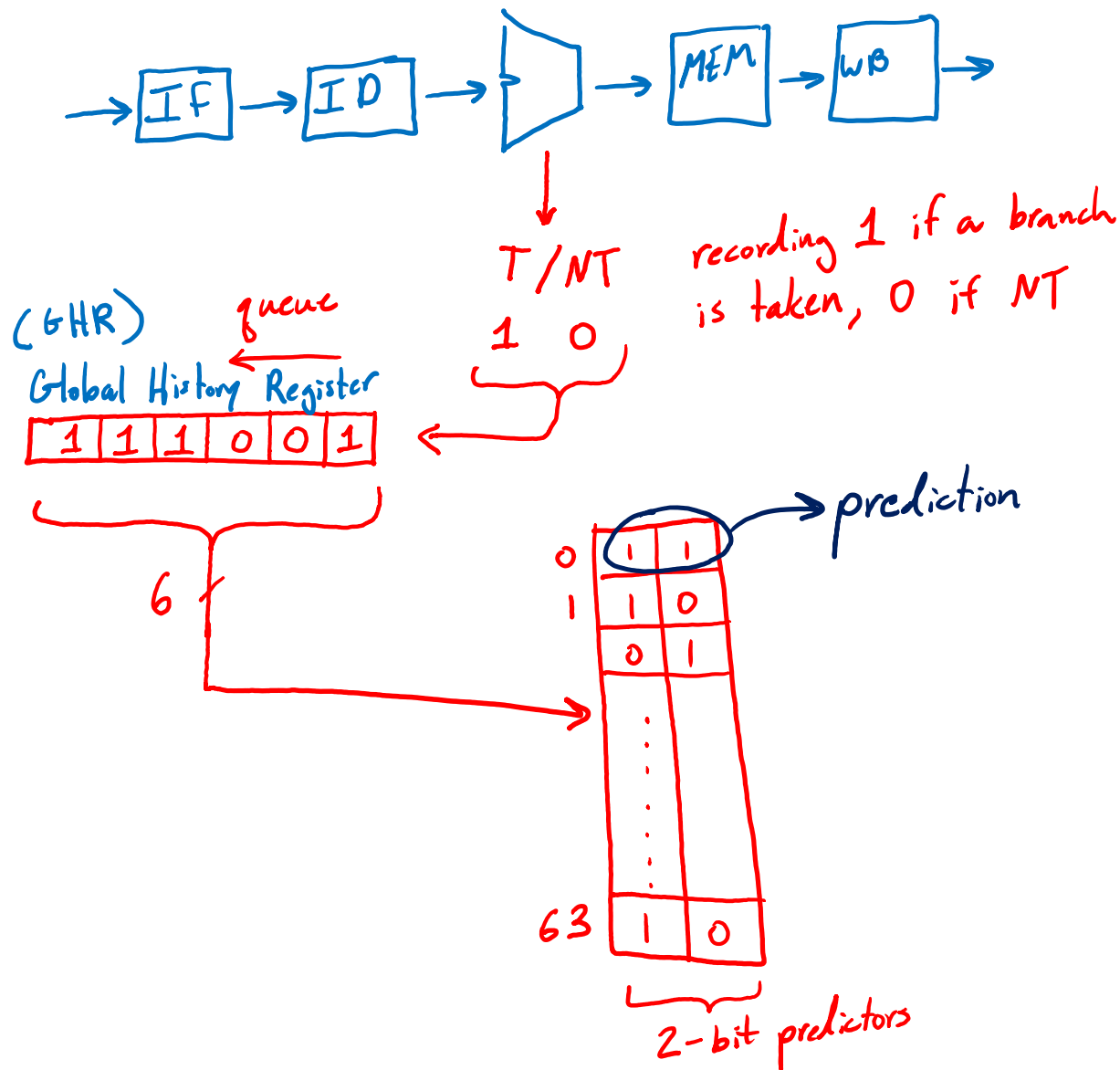
Local Branch Predictor

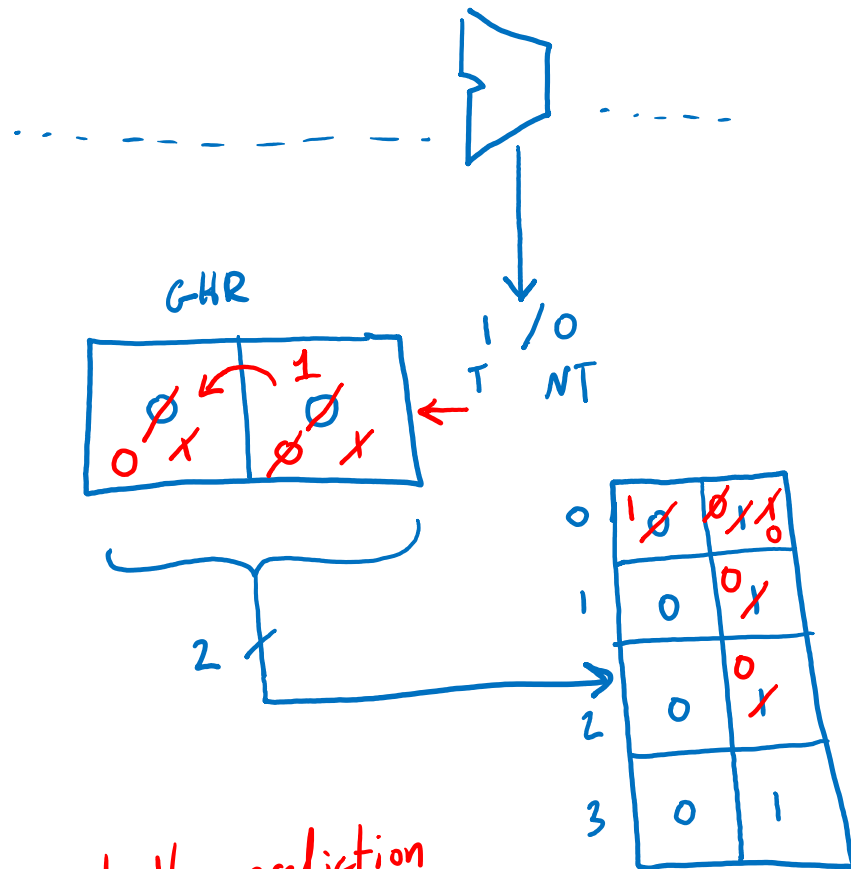




NT 0
 NT 0
 T 1
 NT 0
 NT 0
 T 1
 NT 0

Correlating
 Branch
 Predictor





1. get the prediction
2. update the prediction
3. update the GHR

Prediction	Actual	Correct?
NT	NT	✓
NT	NT	✓
NT	T	X
NT	NT	✓
NT	NT	✓
NT	T	X

Example 1

A program is composed of 25% loads, 25% stores, 30% ALU inst, and 20% branches. Assume 30% of the loads are followed by an instruction that uses the load result. Assume branches are resolved in the ID stage and 40% of the branches are taken (next instruction is always fetched).

What is the average CPI for the program?

$$\begin{aligned} &= \underbrace{.25 \left(\overset{\text{loads}}{\downarrow} .30 \left(\overset{\text{use after load}}{\downarrow} 1 + \overset{\text{stall}}{\downarrow} 1 \right) + \overset{\text{no stall}}{\downarrow} .70 (1) \right)}_{\text{loads}} + \overset{\text{stores}}{\downarrow} .25 (1) + \overset{\text{ALU}}{\downarrow} .30 (1) + \\ &\quad \overset{\text{branches}}{\uparrow} .20 \left(\overset{\text{taken}}{\uparrow} .40 (1 + 1) + \overset{\text{squash}}{\uparrow} .60 (1) \right) \\ &= 1.155 \text{ CPI} \end{aligned}$$